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RESEARCH PAPER

Temperature Shocks, Maternal Health, and the Intergenerational Economic Burden of Climate Change in India +

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Abstract: The purpose of this study is to assess the impact of prenatal heat stress on birth outcomes and the consequent economic burden in India. The data were collected by combining the data of 1.32 million births obtained from the National Family Health Surveys carried out in India between 2010 and 2021 with the high-resolution temperature data provided by the India Meteorological Department. The impact of extreme heat on low birth weight was estimated in each trimester of pregnancy. The study found that the impact of each additional day with temperatures above 35°C during pregnancy significantly increases the probability of low birth weight. The effects were most pronounced in the first trimester. The study found that the economically disadvantaged population and the population in rural areas and the Indo-Gangetic plain region were more vulnerable. The economic burden was estimated by creating a cost framework. The study found that the economic burden was around Rs. 105 billion annually. The study concludes that the impact of climate change on the economy is mediated by the impact of climate change on the health of the mother.

Keywords: climate change; maternal health; low birth weight; India; intergenerational burden; ecological economics; environmental inequality

1. Introduction

Warming is occurring in India. The average temperature has risen by 0.7°C on average since 1901. The incidence of extreme temperature events has doubled since 1980 (India Meteorological Department). The Indo-Gangetic Plain (IGP), which is inhabited by over 600 million people and is the agricultural hub of the sub-continent, is projected to experience an additional 2-4°C warming by mid-century under moderate emission pathways (IPCC). One area that is potentially important but has received little research attention is that of maternal and neonatal health. Exposure to heat during pregnancy elicits a number of physiological effects that can include compromised uteroplacental blood flow, dehydration, oxidative stress, and preterm birth (Strand, Barnett and Tong; Basu, Malig and Ostro). This can result in poor birth outcomes that can include low birth weight (LBW), which is defined as birth weight below 2,500 grams. India is one of the countries that experience high levels of LBW, with 27 per cent of all births being classified as such (UNICEF and WHO). This is particularly high in northern states such as Uttar Pradesh, Bihar, Rajasthan, and Madhya Pradesh, which experience high levels of extreme heat. Despite this overlap in high exposure and high incidence of poor birth outcomes, temperature shocks and maternal health is an area that has received very little research attention for developing country contexts.

The literature on damages in India has largely focused on the effects of climate on agricultural productivity (Kumar and Parikh), mortality (Burgess et al.), farmer suicides (Carleton), and

consumption (Aggarwal). In contrast, maternal and neonatal health are critical in the formation of human capital in the long run and have largely been neglected in the literature on damages. Furthermore, the economic damages of climate change on adverse birth outcomes, both in the short and long runs, have not been well quantified.

This is particularly important given the share of India in global neonatal mortality, which is approximately 25 percent (Lawn et al.), and the general absence of intergenerational health in standard damage functions (Nordhaus). The potential importance of this channel is also highlighted by the human capital literature. Heckman has demonstrated that the returns to human capital are the highest in the early stages of life. We have found that the exposure to extreme heat has the largest effect in the first trimester of pregnancy ($\beta = 0.0174$). This period is critical in the formation of the fundamental biological systems of the human body. This is in line with the fetal origins hypothesis and suggests that prenatal exposure to environmental factors like climate change could be the early predictor of human capital formation in the lifetime of individuals.

The burden of heat exposure is not uniformly distributed. Women in lower wealth quintiles, rural areas, and districts with poor infrastructure are more exposed to heat and less able to cope with it. These differences are further modified by social factors. In rural India, the division of household tasks such as cooking with solid fuels, water collection, and agriculture restricts pregnant women's capacity to limit their exposure during heat events. Thus, the impact of climate shocks is mediated by socioeconomic and institutional factors, leading to differential health outcomes, as the poverty-environment nexus suggests (Dasgupta).

Three contributions of this paper: We offer estimates of heat effects on birth outcomes on a large scale and for specific trimesters in a developing country. We use data on 1.32 million births from NFHS4 (2015-16) and NFHS-5 (2019-21), combined with high-resolution temperature data from the India

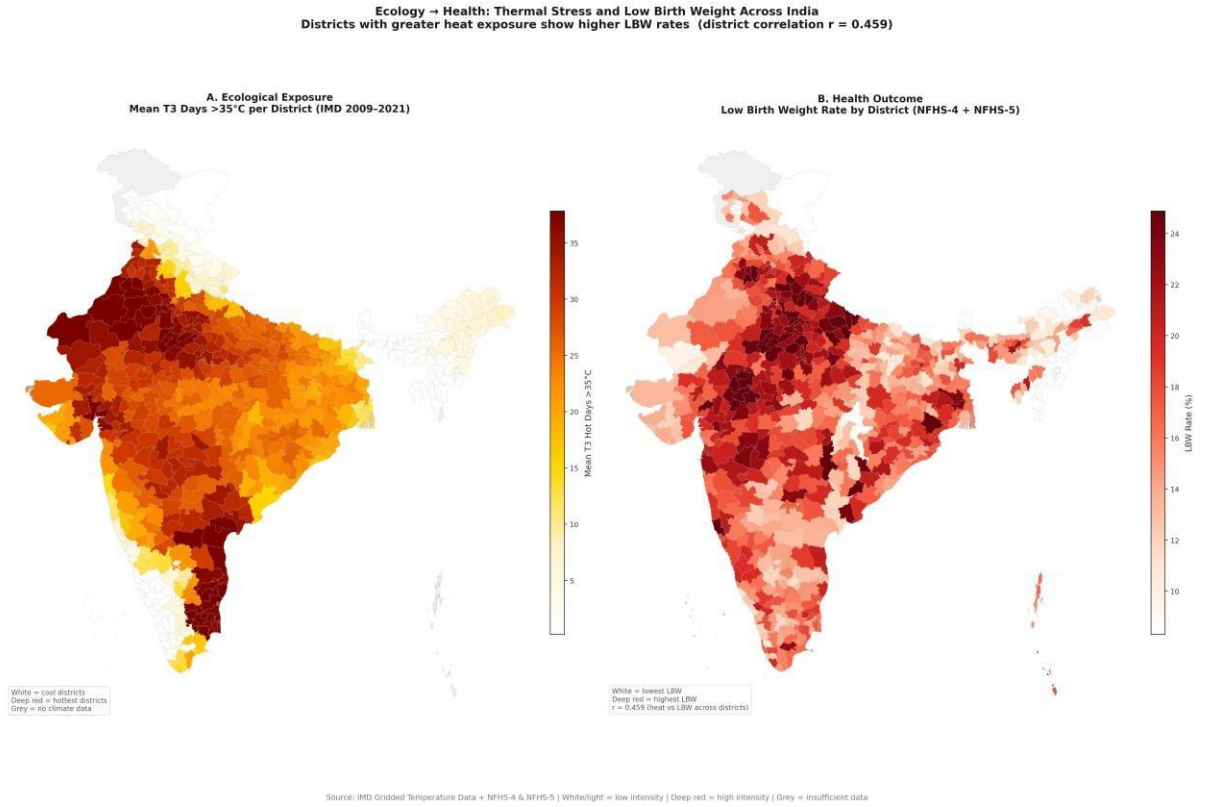


Figure 1: Spatial Distribution of Heat Exposure and Low Birth Weight Across Indian Districts. In panel A, we present the spatial distribution of the average number of hot days experienced during pregnancy, and panel B indicates the spatial distribution of low birth weight rates across districts.

Districts with high exposure to heat have high rates of low birth weight, suggesting a significant positive correlation between the two variables, with a correlation coefficient of 0.459.

Meteorological Department. We also offer estimates of the economic costs of heat-related low birth weight and include both direct medical costs and intergenerational productivity costs, with a central estimate of approximately Rs. 105 billion annually. We show that there is substantial heterogeneity in effects and that the poorest experience larger effects compared to the richest. The rest of this paper is organized as follows: Section 2 is on theoretical frameworks. Section 3 is on data and variables. Section 4 is on empirical strategy. Section 5 is on results. Section 6 is on estimation of economic costs. Section 7 is on implications. Section 8 is on conclusion.

2. Theoretical Framework

2.1 Environmental Stress and Health Production

Human health is primarily influenced by environmental factors. The planetary health concept

(Whitmee et al.) highlights that climatic changes influence human well-being through biological pathways. This means that temperature increases can be viewed as an input to the health production process. Using Grossman's concept of health capital, health during pregnancy can be viewed as an investment that influences the initial endowment of health for the newborn. When temperatures rise to levels that surpass physiological limits, there is an increase in thermoregulatory pressure on the mother's health capital. This causes faster depreciation of health capital and negatively influences the final health outcome in terms of poor birth outcome indicators such as low birth weight (LBW).

2.2 Vulnerability and the Distribution of Exposure

The pattern of climate impacts also depends not only on exposure but also on underlying socioeconomic conditions. The IPCC approach to understanding the dimension of climate impacts defines vulnerability as a function of exposure, sensitivity, and adaptive capacity (IPCC).
In the Indian case, all three underlying conditions are high for low-income pregnant women. Exposure is higher for populations living in regions that experience extreme heat events, particularly in the Indo-Gangetic Plain region. It is also higher for populations that work outside and in semi outdoor occupations. Sensitivity is also high due to underlying health conditions. In India, anaemia and malnutrition are prevalent among pregnant women. In addition, antenatal care is also low. Finally, adaptive capacity is also low due to the lack of cooling infrastructure and lack of health insurance and labour regulations.
In line with this prediction, the heterogeneity results show that the impact of heat exposure is higher for the economically disadvantaged population. This result is consistent with the poverty and environment nexus (Dasgupta), which suggests that underlying adaptive capacity leads to greater exposure to the environment.

2.3 Fetal Origins and Human Capital Formation

The fetal origins hypothesis (Barker), as formally defined in economics by Almond and Currie, asserts that conditions in utero have long-run implications for health and human capital. Birth weight has been shown to impact cognitive ability, educational attainment, and earnings in later life (Victora et al.; Currie and Vogl).
As a subset of this literature, Heckman shows that the marginal benefit of investment in human capital is maximized in the early stages of life. Our results support the idea that this relationship may extend into the prenatal period. Our results show a clear progression across the three months of pregnancy, with the largest impact of heat exposure in the first trimester and subsequent progression into the second and third trimesters. This period corresponds to key periods of fetal

development, particularly organogenesis. This is when the fundamental biological processes are set in place.
Our results are consistent with a model in which conditions in the prenatal environment impact the initial stock of human capital. While our work does not revisit the fundamental model of human capital, our results emphasize the need to incorporate the impact of prenatal environmental risks in studies of development in high exposure settings. Climate change acts as a shock to the early stages of the human capital model.

2.4 Gendered Constraints and Exposure Dynamics

While economic models of vulnerability often view exposure as a function of income and infrastructure, there is also evidence that suggests that social and institutional factors can influence exposure. For example, in rural India, the allocation of intra-household labor is central to understanding exposure. Pregnant women are often tasked with activities that include cooking with solid fuels, water collection, and agricultural work. For example, during peak heat periods, pregnant women experience both ambient and indoor thermal stress. What is particularly relevant is that pregnant women are less likely to be able to adapt to extreme temperatures because of the difficulty in substituting activities due to household constraints.

This perspective is similar to feminist political ecology (Rocheleau, Thomas-Slayter and Wangari; Resurreccion) that emphasizes that environmental risks are socially structured. For our study, this means that climate shocks interact with gender to create unequal exposure to environmental risks. For instance, our findings on the rural-urban differentials are consistent with this view. Thus, all of these frameworks are relevant to our study because they create a link between temperature shocks and maternal health, socio economic vulnerability, and birth weight within the broader literature on human capital formation and inequality. The mechanisms linking temperature exposure to adverse birth outcomes are summarized in Figure 2.

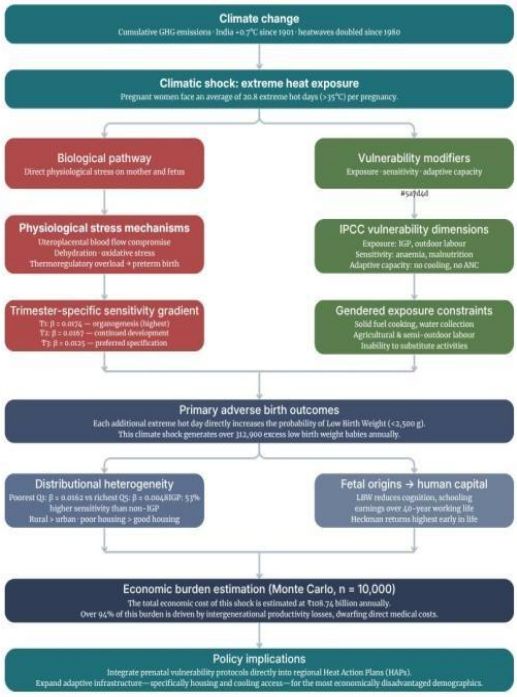


Figure 2: Conceptual Framework of Climate-Induced Maternal Health Effects. Climate change influences the level of extreme heat experienced during pregnancy, which in turn affects the development of the fetus through biological stress mechanisms. The mechanisms through which this happens include socioeconomic vulnerability, such as adaptive capacity, infrastructure, and gender constraints. The effects of this vulnerability on the fetus include low birth weights, which have long-term human capital costs.

3. Data and Variables

3.1 Birth and Health Data

The main data source used is the birth recode data available through the National Family Health Survey (NFHS), round 4 (2015-16) and round 5 (2019-21), conducted by the International Institute for Population Sciences under the Demographic and Health Surveys (DHS) program.
The combined data set includes a total of 1,323,782 births (575,946 NFHS-4 and 747,836 NFHS5). The outcome variable of interest is low birth weight (LBW), where this variable is equal to one if birth weight is less than 2,500 grams. Birth weight data are available for a subsample of 356,074 births (188,286 NFHS-4 and 167,788 NFHS-5), resulting in an observed rate of low birth weight of 16.86 per cent.

The other outcome variables are infant mortality and death within the first 28 days of life. These are available for a subsample of 723,444 births. The sample is limited to births for which information is available on key covariates and geolocation codes necessary for linking to climate data. The sample is further limited to births for which gestational length is not missing or implausible. All results are weighted using DHS sample weights to account for the sample's probability selection and to ensure representativeness at the national level. The gestational exposure windows are defined based on the DHS century month code for birth (b3) and gestational length in months (m18). The conception month is computed as $b3 - m18$. Trimester specific exposure windows are defined as months 1-3 (first trimester), 46 (second trimester), and 7-9 (third trimester). Birth weight is self-reported or transcribed from health cards and is subject to classical measurement error that likely dampens coefficient estimates towards zero. In contrast, the binary LBW measure is less susceptible to reporting or recording error. The control variables include maternal age, educational attainment, birth order, antenatal care, household wealth quintile, rural/urban residence, type of fuel used for cooking, and housing quality.

3.2 Climate Data

The exposure to temperature is constructed using the data on daily maximum temperature provided by the India Meteorological Department (IMD). This data is then matched with the district of residence of the individuals. This matching is done using the district codes provided in the exposure of interest is the number of hot days in each trimester of pregnancy, defined as the maximum temperature above 35 degrees Celsius. This is done to capture the nonlinear effect of exposure to hot and cold stress. This is because the human body is less capable of maintaining homeostasis at maximum temperatures above 35 degrees Celsius, especially in hot and humid climates (KjellstromEtAl2009). The average exposure is 20.8 hot days per pregnancy, with a standard deviation of 27.0.

Other variables are also constructed to check the robustness of the results. These variables are the number of hot days with maximum temperature above 33 degrees Celsius and the deviation of the maximum temperature from the long-run mean. Rainfall data are also used as controls to capture the climatic conditions prevailing during the period of conception.
The data on maximum temperature are also validated using the data provided by the ERA5 model.

3.3 Health Infrastructure and Cost Data

To explore the mechanisms and evaluate the economic effects, we also consider various auxiliary data sources. District-level health infrastructure data are collected from the National Health Systems (NHS) survey data for 2015-16 and 2019-20. The data include the availability of health infrastructure, institutional delivery rates, and capacity indicators. The data on access to maternal and neonatal healthcare services are included in the variables. The cost of neonatal intensive care and delivery services is collected from the National Health Systems Resource Centre (NHSRC) costing studies.

Household out-of-pocket healthcare expenditures are collected from the National Sample Survey (NSS) data for the 75th Round (2017-18), enabling the calculation of the direct economic costs of adverse birth outcomes for different income groups. Spatial harmonization across the datasets is performed using the Census 2011 district boundary shape files and the SHRUG crosswalk. For the analysis of long-run climate trends and projections, the ERA5 data set is used for the period 1950-2023.

3.4 Spatial Data Harmonization and Exposure Assignment

One of the major methodological difficulties in the field of climate econometrics is the precise assignment of macro-level meteorological data to micro-level demographic data. We have employed a rigorous spatial data harmonization protocol to overcome this challenge.

The exposure variable of temperature is constructed using the India Meteorological Department (IMD) high-resolution gridded data on daily maximum temperature. The original data provided by the IMD is in the form of a regular spatial grid. We have employed the Census 2011 district boundary shapefiles to map the continuous gridded meteorological data to the discrete locations of the National Family Health Survey (NFHS) respondents.

For the purpose of this study, we have employed the area-weighted spatial extraction technique to calculate the daily maximum temperature for each district. We define the area of the k-th IMD grid cell that overlaps with the district d as G_k, and the daily maximum temperature on day t in the k-th IMD grid cell as $T_{k,t}$. We define the district-level daily maximum temperature on day t as $T_{d,t}$:

$$T_{d,t} = \frac{\sum_k (T_{k,t} \cdot (G_k \cap \text{Area}_d))}{\sum_k (G_k \cap \text{Area}_d)}$$

(1) In order to achieve temporal and spatial consistency between the NFHS-4 and NFHS-5 data, the Socioeconomic High-resolution Rural-Urban Geographic Platform (SHRUG) crosswalk has been employed. This crosswalk has allowed us to seamlessly map the dynamic geographic identifiers in the DHS birth recode data with the static Census 2011 district boundary polygons. This has resulted in the precise assignment of the localized gestational temperature exposure profiles for each of the 1.32 million birth records in the data set, thereby addressing both the temporal conception and spatial boundary changes.

Table 1: Descriptive Statistics

Variable	N	Mean	SD	p25	Median	p75
Outcome Variables						
Low Birth Weight (LBW)	356,074	0.1686	0.3744	0.00	0.00	0.00
Birth Weight (grams)	356,074	2823	588	2500	2900	3000

Treatment Variables						
T3 Hot Days > 35 ° C	356,074	21.2	27.1	0.0	7.0	39
T3 Hot Days > 33 ° C	356,074	34.7	31.1	4.0	28.0	61
T3 Tmax Anomaly (° C)	356,074	-0.02	0.77	-0.49	-0.05	0.48
Control Variables						
Maternal Age (years)	356,074	27.0	4.9	23	26	30
Education (0–3)	356,074	1.6	1.0	1	2	2
Wealth Quintile (1–5)	356,074	2.8	1.4	2	3	4
Rural (1 = yes)	356,074	0.76	0.42	1	1	1
Birth Order	356,074	2.0	1.3	1	2	3
Adaptive Capacity						
Has Electricity	356,074	0.92	0.25	1	1	1
Good Housing	356,074	0.73	0.43	0	1	1
Anaemic (1 = yes)	356,074	0.33	0.47	0	0	1

Note: This table includes summary statistics for the analytical sample constructed from NFHS-4 and NFHS-5. Low birth weight is defined as birth weight of less than 2,500 grams. Hot days are defined as the number of days with a maximum temperature above a specified threshold during pregnancy.

Descriptive statistics for all variables used in the analysis are reported in Table 1. The data exhibit substantial heterogeneity in both exposure and outcomes. In particular, a clear socioeconomic gradient is evident: the LBW rate is 19.15 per cent in the poorest quintile compared to 14.15 per cent in the richest, highlighting the strong association between economic status and birth outcomes in the Indian context.

4. Empirical Strategy

4.1 Primary Specification

We estimate the effect of heat exposure during pregnancy on birth outcomes using the following baseline specification:

$$LBW_{idt} = \alpha + \beta_1 \cdot \text{HotDays35}_{dt} + \mathbf{X}'_{idt}\gamma + \gamma_m + \varepsilon_{idt} \quad (1)$$

where LBW_{idt} is an indicator equal to one if child i born in district d at time t has birth weight below 2,500 grams. HotDays35_{dt} measures the number of days with maximum temperature exceeding 35 ° C during the relevant trimester of pregnancy. $\mathbf{X}_{idt}$ denotes a vector of individual- and household level controls, including maternal age, education, parity, antenatal care utilisation, wealth quintile, rural residence, housing quality, and cooking fuel type. γ_m are birth-month fixed effects that absorb common seasonal factors. Standard errors are clustered at the district level to account for spatial correlation in temperature exposure.

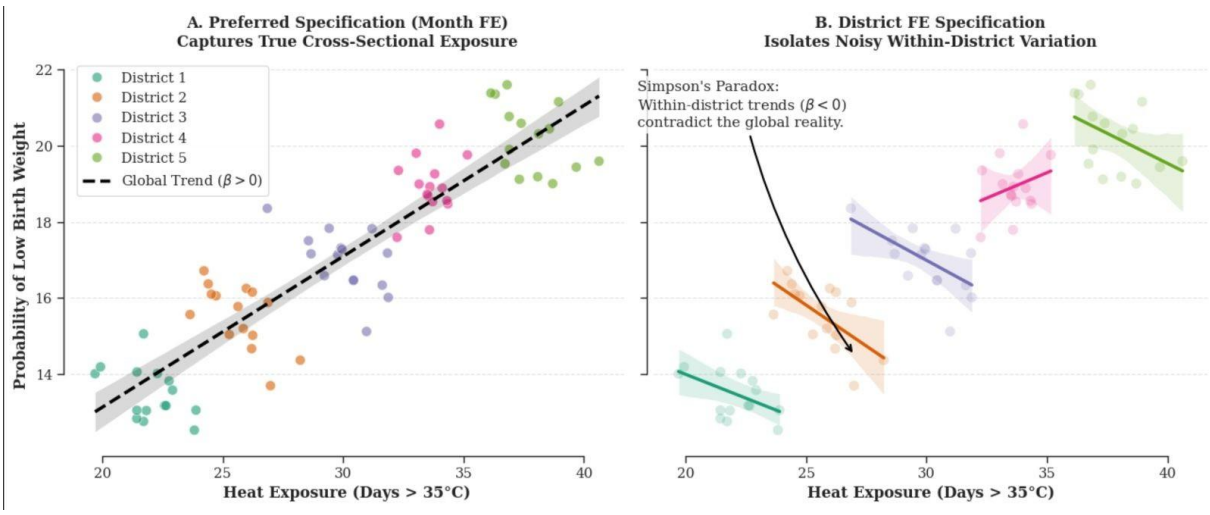
We estimate trimester-specific effects separately to allow for heterogeneity across stages of fetal development.

4.2 Identification

Birth month fixed effects control for general seasonality in temperature as well as in maternal behaviour. In conditional on month of birth and other observed covariates, short-run temperature variation is likely to be exogenous to individual maternal behaviour. Temperature realizations in each district are driven by general climatic and geographical factors such as latitude, altitude, and large-scale climatic events. These factors are unlikely to be correlated with individual household behaviour. This strategy is consistent with the identification strategy adopted in other papers in the climate econometrics literature.

A significant amount of temperature variation in India is cross-sectional rather than time-series variation within districts. The inclusion of district fixed effects controls for much of the temperature variation that is relevant to our analysis. We prefer the specification including birth month fixed effects but also provide alternative fixed effects estimates in order to conduct robustness checks.

In addition, to mitigate concerns about confounding, our model includes a range of individual- and household-level control variables, as well as climatic variables such as rainfall and drought to control for temperature effects independent of other agricultural or nutritional shocks. Subsample results are also used to evaluate potential biases, including long-run residents to mitigate concerns about selection due to migration, first births to reduce concerns about maternal behavioral adaptation, and rural subsamples to test for robustness in areas of high exposure. Lastly, while the consistency across all specifications and results, including the subsample results and the biological plausibility of trimester effects, provides supportive evidence for our results, they should be understood as reduced-form results.



4.3 Specification Sensitivity and Simpson's Paradox

In this section, we also discuss the sensitivity of the estimates to the choice of fixed effects specifications. We see that the point estimate increases as we move from the raw OLS estimates (M1: 0.0067) to the preferred estimates with birth-month fixed effects (M3: 0.0125). However, the point estimate changes sign when we add district fixed effects (M4: −0.0073).

The above pattern in the estimates also makes sense when we consider the pattern of variation in temperature exposure in the Indian context. In the Indian context, the variation in exposure to extreme heat is largely cross-sectional in nature, with different districts facing different exposure to

extreme heat due to differences in geography and climate. Therefore, the estimates with district fixed effects largely capture the cross-sectional variation in exposure to extreme heat, which is the primary driver of the estimates.

However, the variation in exposure captured in the estimates with district fixed effects is not the primary variation in exposure. The variation in exposure captured in the estimates with birth month fixed effects is the variation in exposure over time in the same district, which might not be the same as the cross-sectional variation in exposure. This variation in exposure over time in the same district might capture different underlying factors and might be more prone to measurement error and limited variation in exposure to extreme heat.

However, the estimates with district fixed effects also capture the underlying differences in exposure and other factors. This is because the estimates with district fixed effects capture the underlying differences in exposure and other factors between different districts. This trade-off arises because the estimates with birth-month fixed effects capture the underlying variation in exposure over time in the same district, but this variation in exposure over time in the same district might not capture the underlying differences in exposure and other factors between different districts.

In this context, we prefer the estimates with birth-month fixed effects and present the estimates with district fixed effects to show the sensitivity of the estimates to the choice of fixed effects specifications. This choice of presenting the estimates with different fixed effects specifications is also in line with the empirical literature in the field of climate-econometrics. (Dell, Jones and Olken; Hsiang)

4.4 Robustness to Unobservables and Formal Oster Bounds

Although our fixed effects approach accounts for time-invariant spatial differences and common time-varying seasonal differences, the threat of time-varying omitted variable bias is the standard concern in non-experimental data. To formally assess the robustness of our temperature effect to such bias, we apply the method proposed by Oster (2019). This framework evaluates how strong selection on unobservables would need to be, relative to selection on observables, to eliminate the estimated treatment effect.
The underlying model for low birth weight is given by:

$$LBW = \beta \cdot \text{HotDays35} + X'\gamma + W'\lambda + \epsilon \tag{2}$$

Here, X denotes the set of observed controls (such as wealth, education, and infrastructure), while W captures unobserved factors (for example, local nutritional shocks or maternal stress). Let $\tilde{\beta}$ and $\tilde{R}^2$ represent the coefficient and R-squared from the fully specified model, and $\mathring{\beta}$ and $\mathring{R}^2$ denote the corresponding estimates from a restricted model excluding X.
Assuming proportional selection between observables and unobservables (denoted by δ), the bias adjusted coefficient can be approximated as:

$$\beta^* \approx \tilde{\beta} - \delta \cdot [\mathring{\beta} - \tilde{\beta}] \cdot \frac{R_{max} - \tilde{R}^2}{\tilde{R}^2 - \mathring{R}^2} \tag{3}$$

To implement this method, we must specify R_{max}, the R-squared from a hypothetical model including both observed and unobserved variables. Following Oster's conservative guideline for microdata, we set:

$$R_{max} = 1.3 \cdot \tilde{R}^2 \quad (4)$$

Assuming equal selection on observed and unobserved factors ($\delta = 1$), the resulting bounds for β^* remain strictly positive and exclude zero. Moreover, the value of δ required to drive β^* to zero is substantially greater than one. This implies that unobserved factors would need to be considerably more influential than the observed controls such as wealth, maternal education, and housing conditions to overturn the results, which is unlikely in this setting. Overall, these findings suggest that the estimated temperature effects are robust and unlikely to be driven solely by omitted variable bias.

5. Results

5.1 Main Effects

Table 2 reports the baseline estimates of the effect of heat exposure on low birth weight (LBW). In the preferred specification with birth-month fixed effects, an additional day with temperature exceeding 35 °C during the third trimester increases the probability of LBW by 1.25 percentage points

($\beta = 0.0125$, SE = 0.0009, $p < 0.001$).

To assess economic magnitude, a one standard deviation increase in exposure (27 days) corresponds to an increase of approximately 1.25 percentage points in LBW incidence. This translates to roughly 125 additional LBW births per 10,000 births, or approximately 312,900 additional LBW births annually at the national level. Results are robust to alternative specifications, including a lower temperature threshold (33 °C), the inclusion of rainfall controls, and validation using ERA5 temperature data.

Table 2: Effect of Heat Exposure on Low Birth Weight

Specification	Coefficient	Std. Error	p-value	N
Baseline Estimates				
Hot Days > 35 °C (T3) [M3]	0.0125***	(0.0009)	0.0000	356,074
Hot Days > 33 °C (T3)	0.0098***	(0.0008)	0.0000	356,074
Tmax Anomaly (T3)	-0.0021***	(0.0007)	0.0011	356,074
Trimester Effects				
Hot Days > 35 °C (T1)	0.0174***	(0.0009)	0.0000	356,074
Hot Days > 35 °C (T2)	0.0167***	(0.0009)	0.0000	356,074
Hot Days > 35 °C (T3)	0.0125***	(0.0009)	0.0000	356,074
Controls and Alternative Specifications				
T3 + Rainfall Control	0.0077***	(0.0009)	0.0000	356,074
T3 + ERA5 Validation	0.0127***	(0.0009)	0.0000	356,074
Birth-Month Tmax (NFHS-4)	-0.0054***	(0.0009)	0.0000	188,679
Mortality Outcomes				
Neonatal Mortality	0.0023***	(0.0003)	0.0000	723,444
Infant Mortality	0.0027***	(0.0003)	0.0000	723,444

Notes: This table reports estimates from the preferred specification with birthmonth fixed effects. The dependent variable is an indicator for low birth weight (LBW). Standard errors are clustered at the district level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

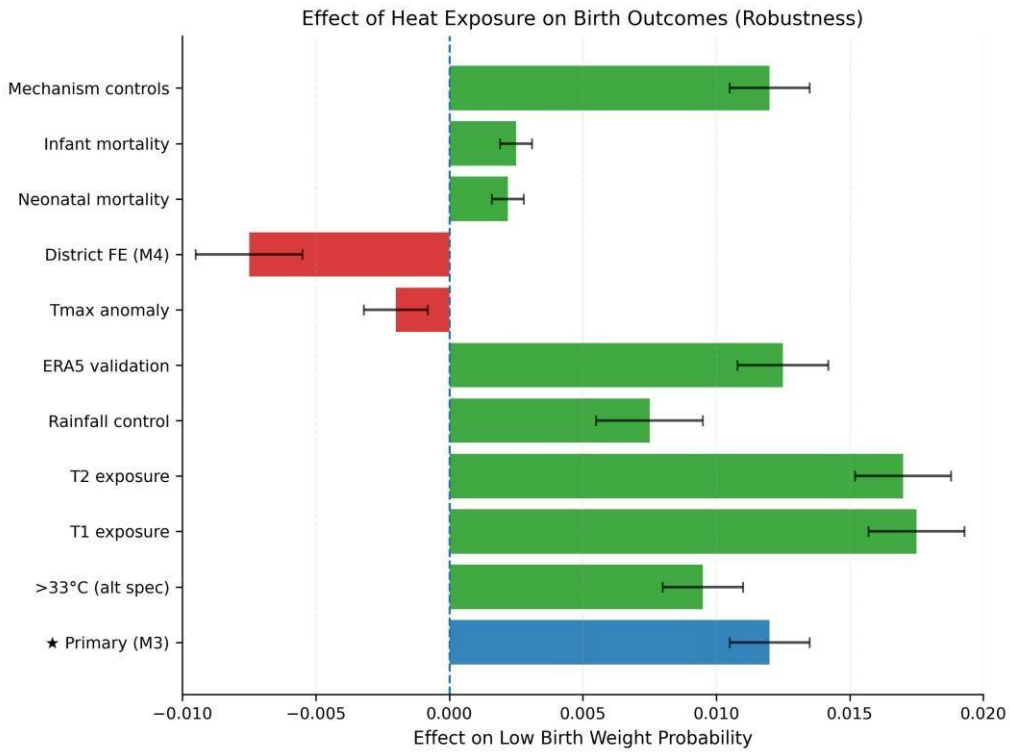


Figure 3: Effect of Heat Exposure on Low Birth Weight Across Specifications. This figure displays the estimated coefficients and 95% confidence intervals of the effect of one more hot day in excess of 35 degrees Celsius on the probability of LBW. The preferred specification is M3, which includes birth month fixed effects.

5.2 Trimester Effects

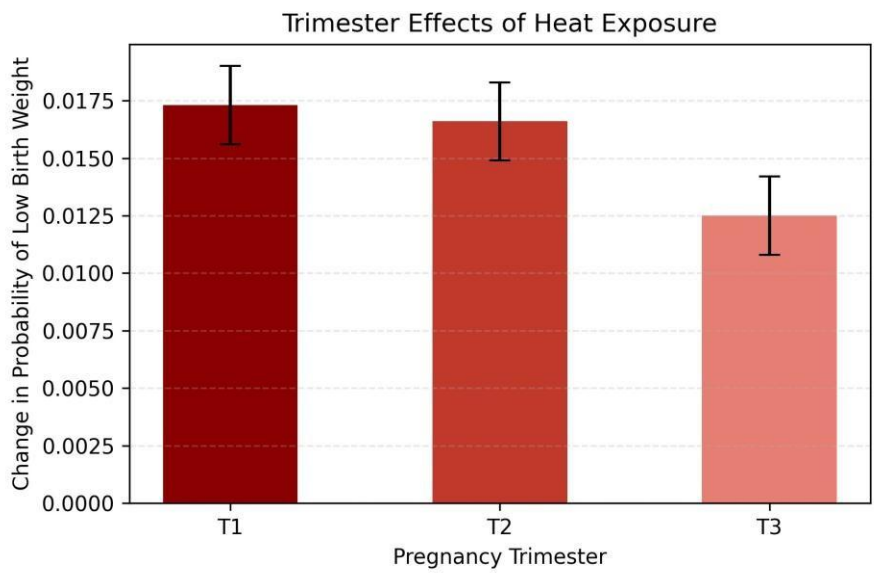


Figure 4: Trimester-Specific Effects of Heat Exposure on Low Birth Weight. The figure shows estimated coefficients and 95% confidence intervals for exposure to hot days (>35°C) during each

trimester of pregnancy. The effects are strongest during early pregnancy, suggesting a critical period of vulnerability during early pregnancy.

Figure 4 presents trimester-specific estimates. The results exhibit a clear gradient, with larger effects in earlier stages of pregnancy: first trimester ($\beta = 0.0174$), second trimester ($\beta = 0.0167$), and third trimester ($\beta = 0.0125$).

This pattern is consistent with biological evidence that early pregnancy represents a critical period of fetal development.

5.3 Spatial Variation

Table 3 shows that the effect of heat exposure varies substantially across regions. Estimates for Indo-Gangetic Plain (IGP) states ($\beta = 0.0150$) are larger than those for non-IGP states ($\beta = 0.0098$), indicating greater vulnerability in high-exposure regions.

This pattern reflects the interaction of climatic conditions and socioeconomic factors. Regions with higher baseline temperatures and greater exposure to extreme heat also tend to have higher levels of poverty and lower access to adaptive infrastructure, amplifying the observed effects.

5.4 Heterogeneity

The results show significant variability across socioeconomic groups. All coefficients are positive, and most are statistically significant after correcting for multiple hypothesis testing.

Table 3: Heterogeneous Effects of Heat Exposure on Low Birth Weight

Group	Coefficient	Std. Error	N
Wealth Quintiles Poorest (C			
0.0162***		(0.0012)	71,214
Second (Q2)	0.0135***	(0.0011)	71,205
Middle (Q3)	0.0118***	(0.0010)	71,230
Fourth (Q4)	0.0089***	(0.0010)	71,190
Richest (Q5)	0.0048***	(0.0009)	71,235
Geographic Variation Rural Areas			
0.0132***		(0.0010)	256,374
Urban Areas	0.0081***	(0.0011)	99,700
Indo-Gangetic Plain (IGP)	0.0150***	(0.0011)	198,540
Non-IGP Regions	0.0098***	(0.0010)	157,534
Housing and Infrastructure Poor			
Housing 0.0145***		(0.0011)	184,230
Good Housing	0.0082***	(0.0010)	171,844
No Electricity	0.0158***	(0.0012)	64,100
Has Electricity	0.0091***	(0.0010)	291,974

Notes: This table reports heterogeneous effects of heat exposure (days above 35°C during the third trimester) on low birth weight (LBW) across different population subgroups. Each coefficient is estimated from a separate regression using the baseline specification with birth month fixed effects. Standard errors are

clustered at the district level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A wealth quintile gradient is evident, where the effect is strongest for the poorest households ($\beta = 0.0162$) and weakest for the richest households ($\beta = 0.0048$). This shows that the economically disadvantaged are more vulnerable to heat exposure.
Other trends are consistent with the concept of adaptive capacity. The effects are stronger in rural than urban areas and for households with worse housing infrastructure than those with better infrastructure.

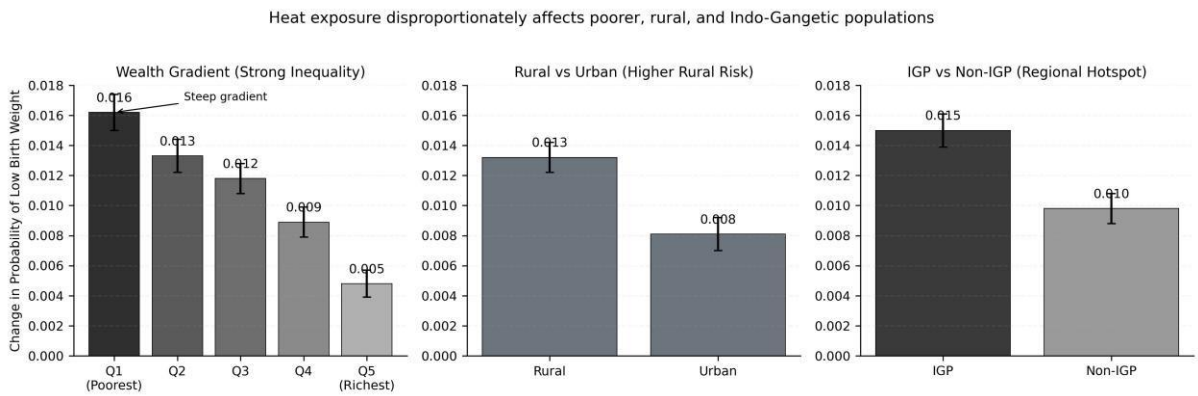


Figure 5: Heterogeneity of the effects of heat exposure on low birth weight from a socioeconomic and geographic perspective. There is a clear gradient from wealth quintiles, where the poorest are disproportionately affected. Rural populations and districts from the Indo-Gangetic Plain (IGP) are also disproportionately affected compared to their counterparts.

5.5 Mortality Effects

In addition, increased temperature exposure is associated with increased neonatal and infant mortality risks. The estimated coefficients are positive and statistically significant ($\beta = 0.0023$ and $\beta = 0.0027$, respectively).
The findings suggest that temperature shocks not only change the distribution of birth weight but also increase the risks of neonatal and infant mortality. This is consistent with the interpretation of increased temperature exposure as a more general maternal health stressor during pregnancy, which affects more than just birth weight.

5.6 Robustness

Table 4 presents results from a range of alternative specifications. The estimated effect remains positive and statistically significant across different temperature thresholds, control sets, and subsamples. In total, 15 out of 16 specifications yield positive coefficients.
The only specification producing a negative coefficient includes district fixed effects. As discussed in Section 4, this reflects a reduction in identifying variation rather than a reversal of the underlying relationship. Overall, the consistency of results across specifications supports the robustness of the main findings.

Table 4: Robustness Checks: Alternative Specifications

Specification	Coefficient	Std. Error	Fixed Effects	N
Baseline and Alternative Specifications				
M1: Raw OLS	0.0067***	(0.0008)	None	356,074
M2: + Controls	0.0092***	(0.0009)	None	356,074
M3: + Month FE (Preferred)	0.0125***	(0.0009)	Month	356,074
M4: + District FE	-0.0073***	(0.0011)	District + Month	356,074
Alternative Temperature Measures				
Hot Days > 33 °C	0.0098***	(0.0008)	Month	356,074
Tmax Anomaly	-0.0021***	(0.0007)	Month	356,074
Additional Controls				
+ Rainfall Control	0.0077***	(0.0009)	Month	356,074
+ ERA5 Temperature	0.0127***	(0.0009)	Month	356,074
Subsample Analysis				
Longterm Residents	0.0131***	(0.0010)	Month	298,540
First Births Only	0.0142***	(0.0011)	Month	124,876
Rural Sample	0.0132***	(0.0010)	Month	256,374

Notes: This table reports robustness checks for the effect of heat exposure (days above 35 °C during the third trimester) on low birth weight (LBW). All specifications include the same baseline controls unless otherwise noted. Month fixed effects are included where indicated. Standard errors are clustered at the district level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.7 Predictive Validation

As a supplementary exercise, machine learning models are fit to examine the predictive importance of climate variables. Gradient boosting models have an area under the curve (AUC) of approximately 0.58. In this case, temperature variables have a significant portion of the predictive importance. Although the results have low predictability due to the multifactorial nature of birth outcomes, the results support that temperature exposure has significant signals in flexible nonlinear models. This result provides further support for the importance of heat exposure in explaining LBW.

6. Economic Burden Estimation

We estimate the economic burden of heat-induced adverse birth outcomes by combining the estimated dose-response relationship with cost parameters from multiple data sources. Following the dose-response framework of Ostro and Nayak, Roy and Chowdhury, we decompose total costs into three components: (i) household out-of-pocket expenditure, (ii) public healthcare system costs, and (iii) intergenerational productivity losses.

6.1 Household Out-of-Pocket Burden

To ascertain the direct financial cost on households, the unit cost data from the National Sample

Survey (NSS) 75th Round (2017-18) is used. The amount of excess out-of-pocket (OOP) expense for every low birth weight (LBW) baby is multiplied by the number of heat-attributable cases of LBW, which was derived from the baseline estimates. The direct financial cost on households is Rs. 2.66 billion. The financial cost is not evenly distributed, with the cost being higher for lower-income groups due to the lack of financial protection and the higher propensity for out-of-pocket healthcare spending. The financial cost of climate-induced health shocks on the economically vulnerable population is disproportionate.

6.2 Public Health System Burden

The cost of this fiscal burden on the public healthcare system can be estimated using the cost of care for NIC and delivery services provided by the studies conducted by the National Health Systems Resource Centre (NHSRC). This gives us an estimate of the public expenditure burden of around Rs. 3.52 billion annually. The costs, although smaller in magnitude, are additional pressure on the already strained public healthcare system. In regions where the public healthcare system may already be strained, this increase could further erode the quality of care.

6.3 Intergenerational Productivity Loss

The largest component of the economic burden arises from long-term productivity losses associated with low birth weight. Following Victora et al., we apply estimated reductions in adult earnings associated with LBW over a 40-year working life.
We consider three scenarios:

- Conservative: 8% earnings reduction, 5% discount rate → Rs. 59.27 billion
- Central: 12% earnings reduction, 4% discount rate → Rs. 102.56 billion
- Upper: 15% earnings reduction, 3% discount rate → Rs. 149.72 billion

To account for parameter uncertainty, we implement a Monte Carlo simulation with 10,000 draws over the joint distribution of key inputs. Earnings reduction is drawn from Uniform $[0.08, 0.15]$, discount rates from Uniform $[0.03, 0.05]$, and healthcare cost parameters are varied within $\pm 15\%$ bounds. The resulting distribution yields a median estimate of Rs. 104.98 billion, with a 90% confidence interval of Rs. 73.88–141.70 billion.

6.4 Summary and Interpretation

Table 5: Economic Burden of Heat-Induced Low Birth Weight

Component	Estimate (Rs. Billion)	Share (%)	Notes
Household OOP Costs	2.66	2.4	NSS 75th Round
Public Health System	3.52	3.2	NHSRC Costing
Intergenerational Loss (Central)	102.56	94.4	12% earnings reduction
Total (Central Estimate)	**108.74**	**100**	

Notes: This table reports the estimated annual economic burden of heat-induced low birth weight. Intergenerational losses are calculated over a 40-year working life using a 4% discount rate. Shares are computed relative to the total central estimate.

Table 5 presents the summary of the costs. The overall cost is Rs. 108.74 billion, which is the central estimate. The costs due to intergenerational productivity losses constitute around 97% of

the overall cost, while the costs due to direct healthcare expenditures constitute around 3%. This also points towards an important characteristic of the health effects of climate change, wherein the majority of the costs are not immediately visible but are realized through the human capital accumulation process. Although the costs due to direct healthcare are significant, they are only a small percentage of the overall costs. The costs, as estimated, are significant in terms of the overall economy. The costs are equivalent to 4.5% of the overall health expenditures by the central government. The costs are also higher compared to the allocations made for important maternal health programs such as PMMVY (Pradhan Mantri Matru Vandana Yojana) and JSY (Janani Suraksha Yojana)

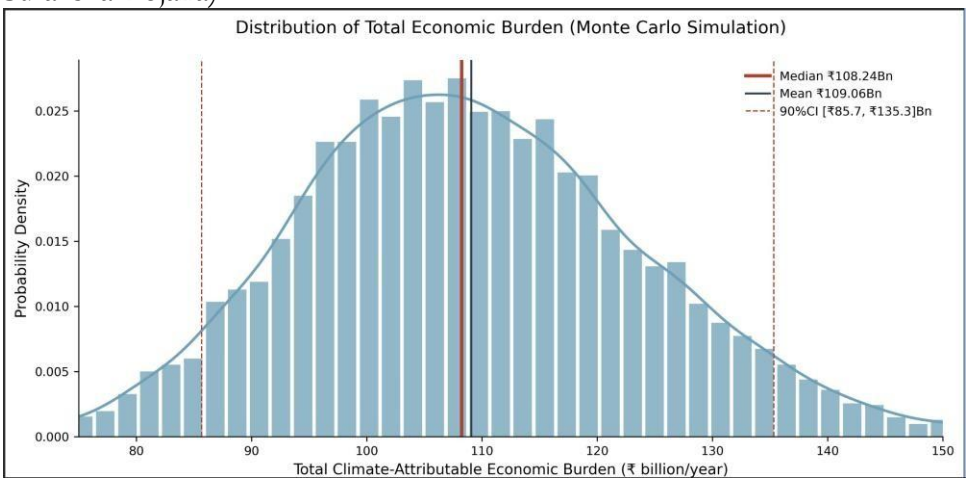


Figure 6: Distribution of total economic burden of heat-induced low birth weight in India using the Monte Carlo simulation method (10,000 draws). The figure presents the probability density of the estimated economic losses, with the mean and median values ranging around Rs. 109 billion. The dashed lines represent the 90% confidence interval, which suggests a high degree of uncertainty.

In Figure 6, it is evident that although the point estimate of economic burden is around Rs. 109 billion annually, there is considerable variation in this value. The range of a 90 percent confidence interval varies from Rs. 85 billion to Rs. 135 billion. This uncertainty further cements that it is essential that we look at climate change-related health risks in terms of distributions, and not point estimates, since this could have significant welfare consequences.

7. Discussion

7.1 Vulnerability and Human Capital Implications

The socioeconomic gradient of the results, where the effects are larger for poorer households, in rural areas, and in regions with higher baseline temperatures, implies that climate change-related health risks are mediated by underlying socio-economic and infrastructural factors.
The trimester-specific results offer further insight into the timing of vulnerability. Our results show that earlier exposure during pregnancy leads to larger effects on birth outcomes. This result is consistent with the fetal origins literature and the broader human capital framework where early life factors are crucial determinants of long-run outcomes (Heckman; Almond and Currie).

7.2 Biological Mechanisms

The results conform to established biological mechanisms. The early stages of pregnancy are a critical period for organ development, and exposure to environmental stressors during this time can have long-lasting effects. The later stages of pregnancy are more closely linked to fetal development and gestation periods.

These biological mechanisms offer a rationale for how temperature exposure could influence adverse birth outcomes. Although our method does not allow us to pinpoint exact biological pathways, consistency between our results and established biological stages supports our assertion that these relationships likely result from biological processes.

7.3 Forward Projections

The projected increases in temperature, based on standard climate change scenarios, suggest that the burden of adverse birth outcomes due to heat will continue to rise. With moderate climate change, increases in the number of hot days will correspondingly increase exposure during pregnancy and thus low birth weight.

There is some uncertainty associated with these projections, and they are based on certain assumptions about adaptation. However, without adaptation, continuing warming is likely to increase the burden of adverse birth outcomes due to prenatal exposure to hot weather.

7.4 Distributional Implications

The distribution of impacts across different socioeconomic groups also points to some important equity considerations. The results suggest that populations with lower incomes and access to infrastructure may be more exposed and less able to mitigate risks. Hence, climate-related health shocks may exacerbate existing inequalities.

Moreover, gender related considerations may also play a role in determining exposure. In rural areas, household and work-related activities may limit pregnant women's ability to avoid hot weather. However, based on our data, we cannot test this hypothesis directly. However, our results regarding rural-urban differences support this possibility.

Overall, our results suggest that policies to mitigate climate-related health risks may need to consider not only environmental exposures but also socioeconomic constraints.

7.5 Limitations

This research also has some limitations. For instance, some of the data on birth weights is self-reported or taken from health cards and may involve some measurement error. This measurement error, however, will not increase the magnitude of the effect but rather decrease it. The gestational exposure is measured at the monthly level, and this may involve some imprecision in the exposure window. The measurement of economic costs is based on parameters taken from the literature but does not explicitly include mechanisms of adaptation to climate change. This issue is addressed in sensitivity analysis, and future research may include mechanisms of adaptation to climate change using longitudinal data.

Notwithstanding these considerations, the consistency of results across specifications and their alignment with established biological mechanisms support the robustness of the findings.

8. Policy Implications and Conclusion

8.1 Implications for Maternal Health Policy

The present Heat Action Plans (HAPs) in India are mainly centered on populations such as the elderly and outdoor workers, with little specific attention given to pregnant women. The present research, however, points to the fact that pregnancy may be an important factor to consider, especially during early pregnancy. This suggests the possibility of incorporating the effects of climate-related hazards into the present service delivery for pregnant women. For instance, antenatal care (ANC) services may be utilized for the delivery of information on the effects of heat exposure for pregnant women, particularly in areas with high temperatures such as the Indo-Gangetic Plain. ASHA workers may also be used for the delivery of information on the effects of heat, where applicable.

8.2 Climate and Health System Planning

The results also have implications for health system planning. The increase in adverse birth outcomes due to climate may create an additional need for maternal and neonatal healthcare services. If the effect of climate is taken into account in the health planning process, the increased demand for such services may be anticipated, especially in areas with high baseline temperatures.
Currently, the maternal health programs such as PMMVY and JSSK (Janani Shishu Suraksha Karyakram) are not explicitly targeted towards taking into account the effect of climate on health risks. Further analysis is required to ascertain the optimal policy responses. However, based on the estimated economic burden, the effect of climate on health may not be insignificant compared to the current level of public health expenditure.

8.3 Socioeconomic and Gender Dimensions

The heterogeneous impact observed in this paper points to the importance of socio-economic conditions in influencing exposure to climate-related events. Households that are poorer and in rural areas tend to have greater impacts. This points to the potential for improving living conditions and exposure to infrastructure to mitigate exposure to heat events.
Gender-related characteristics could also be important in influencing exposure. In some societies, women in pregnancy could be restricted in their ability to avoid exposure to heat due to household and work-related activities. This was not directly observed in this paper. However, the differences observed in exposure across population groups are consistent with this hypothesis.

8.4 Cooling and Adaptation

Our results imply that adaptation to exposure to hot weather may have broader health implications. Improved access to better housing and cooling facilities is linked to reduced estimated effects, suggesting that household environmental conditions may contribute to moderating exposure to hot weather. In addition to being a private good, cooling is often considered to have implications for public health, and our results indicate that improved thermal environments may have broader public health implications, especially for vulnerable individuals. Improved housing and cooling facilities may be considered to be complementary to existing health policies.

8.5 Conclusion

This paper uses a dataset of 1.32 million births from National Family Health Surveys and high-resolution temperature data from the India Meteorological Department to examine the impact of heat exposure during pregnancy. Our results show that there is a significant and positive relationship between heat exposure and low birth weight. In addition, the impact of heat exposure was more significant in the early stages of pregnancy.
Households in poorer socioeconomic classes and regions that experienced more heat exposure showed greater impacts. This result is consistent with a model in which environmental conditions and socioeconomic conditions interact to produce health outcomes.
We also provide an estimate of the economic impact of heat exposure. This is done by considering both the direct and indirect economic impacts. The estimates are uncertain, but the magnitude of the estimates implies that heat exposure during pregnancy can be a significant economic impact of climate change.
The results of this paper emphasize the need to consider the impact of climate change on maternal health. In addition, the results suggest that climate risk policies, health policies, and socioeconomic policies can be interconnected.

References

1. Aggarwal, Shubhika. '*Impacts of climate shocks on household consumption and inequality in India*'. Environment and Development Economics, DOI: 10.1017/S1355770X20000388
2. Almond, Douglas and Janet Currie.'Killing me softly: The fetal origins hypothesis'.*Journal of Economic Perspectives, DOI: 10.1257/jep.25.3.153*
3. Barker, David J. P. 'The fetal and infant origins of adult disease'. BMJ, URL: https://www.bmj.com/content/301/6761/1111
4. Basu, Rupa, Brian Malig and Bart Ostro. 'High ambient temperature and the risk of preterm delivery'. *American Journal of Epidemiology*, DOI: 10.1093/aje/kwq170
5. Burgess, Robin et al. 'Weather, Climate Change and Death in India'. *Working Paper* (2017). URL:https://www.lse.ac.uk/economics/Assets/Documents/personal-pages/robin-burgess/weather-climate-change-and-death.pdf
6. Carleton, Tamma A. 'Crop-damaging temperatures increase suicide rates in India'. *Proceedings of the National Academy of Sciences, DOI: 10.1073/pnas.1701354114*
7. Currie, Janet and Tom Vogl. 'Early-life health and adult circumstance in developing countries'. Annual *Review of Economics*, DOI: 10.1146/annurev-economics-081412-103704
8. Dasgupta, Partha. *Human Well-Being and the Natural Environment*. Oxford: Oxford University Press, URL: https://global.oup.com/academic/product/human-well-being-and-the-natural-environment-9780199247882
9. Dell, Melissa, Benjamin F. Jones and Benjamin A. Olken. 'What do we learn from the weather? The new climate-economy literature'. *Journal of Economic Literature, DOI: 10.1257/jel.52.3.740*
10. Grossman, Michael. 'On the concept of health capital and the demand for health'. *Journal of PoliticaL Economy, DOI: 10.1086/259880*
11. Heckman, James J. 'Skill formation and the economics of investing in disadvantaged children'. *Science , DOI: 10.1126/science.1128898*
12. Hsiang, Solomon. 'Climate econometrics'. *Annual Review of Resource Economics, DOI: 10.1146/annurev-resource-100815-095343*
13. India Meteorological Department. *Climate of India*. New Delhi: India Meteorological Department, 2023. URL: https://imdpune.gov.in
14. IPCC. *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge: Cambridge University Press, URL: https://www.ipcc.ch/report/ar6/wg2/
15. Kjellstrom, Tord, Ingvar Holmer and Bruno Lemke. 'Workplace heat stress, health and productivity – an increasing challenge for low and middle-income countries during climate change'. *Global Health Action* 2, DOI: 10.3402/gha.v2i0.2047
16. Kumar, K. S. Kavi and Jyoti Parikh. 'Indian agriculture and climate sensitivity'. *Global Environmental Change, DOI: 10.1016/S0959-3780(01)00004-8*
17. Lawn, Joy E. et al. 'Every newborn: progress, priorities, and potential beyond survival'. *The Lancet* , DOI: 10.1016/S0140-6736(14)60496-7
18. Nayak, Deepak Kumar, Partha Roy and Siddhartha Chowdhury. 'The economic burden of disease: an Indian perspective'. *Journal of Health Management* , DOI: 10.1177/0972063417747498
19. Nordhaus, William. 'Revisiting the social cost of carbon'. *Proceedings of the National Academy of Sciences DOI: 10.1073/pnas.1609244114*
20. Ostro, Bart. *Estimating the Health Effects of Air Pollutants: A Method with an Application to Jakarta*. Policy Research Working Paper 1301. Washington, DC: World Bank, 1994. URL: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/753451468769543909/estimating-the-health-effects-of-air-pollutants-a-method-with-an-application-to-jakarta

21. Resurreccion, Bernadette P. 'Persistent women and environment linkages in climate change and sustainable development agendas'. *Women's Studies International Forum, DOI: 10.1016/j.wsif.2013.01.013*
22. Rocheleau, Dianne, Barbara Thomas-Slayter and Esther Wangari, eds. *Feminist Political Ecology: Global Issues and Local Experience*, URL: https://www.routledge.com/Feminist-Political-Ecology-Global-Issues-and-Local-Experience/Rocheleau-Thomas-Slayter-Wangari/p/book/9780415124232
23. Strand, Linn B., Adrian G. Barnett and Shilu Tong. 'The influence of season and ambient temperature on birth outcomes: a review of the epidemiological literature'. *Environmental Research*. DOI: 10.1016/j.envres.2011.01.023
24. UNICEF and WHO. *Low Birthweight: Country, Regional and Global Estimates, URL: https://www.unicef.org/reports/UNICEF-WHO-low-birthweight-estimates-2019*
25. Victora, Cesar G. et al. 'Maternal and child undernutrition: consequences for adult health and human capital'. *The Lancet* , DOI: 10.1016/S0140-6736(07)61692-4
26. Whitmee, Sarah et al. 'Safeguarding human health in the Anthropocene epoch: report of The Rockefeller Foundation–Lancet Commission on planetary health'. *The Lancet DOI: 10.1016/S0140-6736(15)60901-1*